

Theory of Automata

CS 3005

Lecture 21

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FAST NUCES CFD

CFG = PDA

- We are now able to prove that the set of all languages accepted by PDAs is the same as the set of all languages generated by CFGs.
- We can prove this in two steps.

Example

- Consider the following CFG in CNF:

$$S \rightarrow SB \mid AB$$

$$A \rightarrow CC$$

$$B \rightarrow b$$

$$C \rightarrow a$$

- We now propose the following nondeterministic PDA where the STACK alphabet is

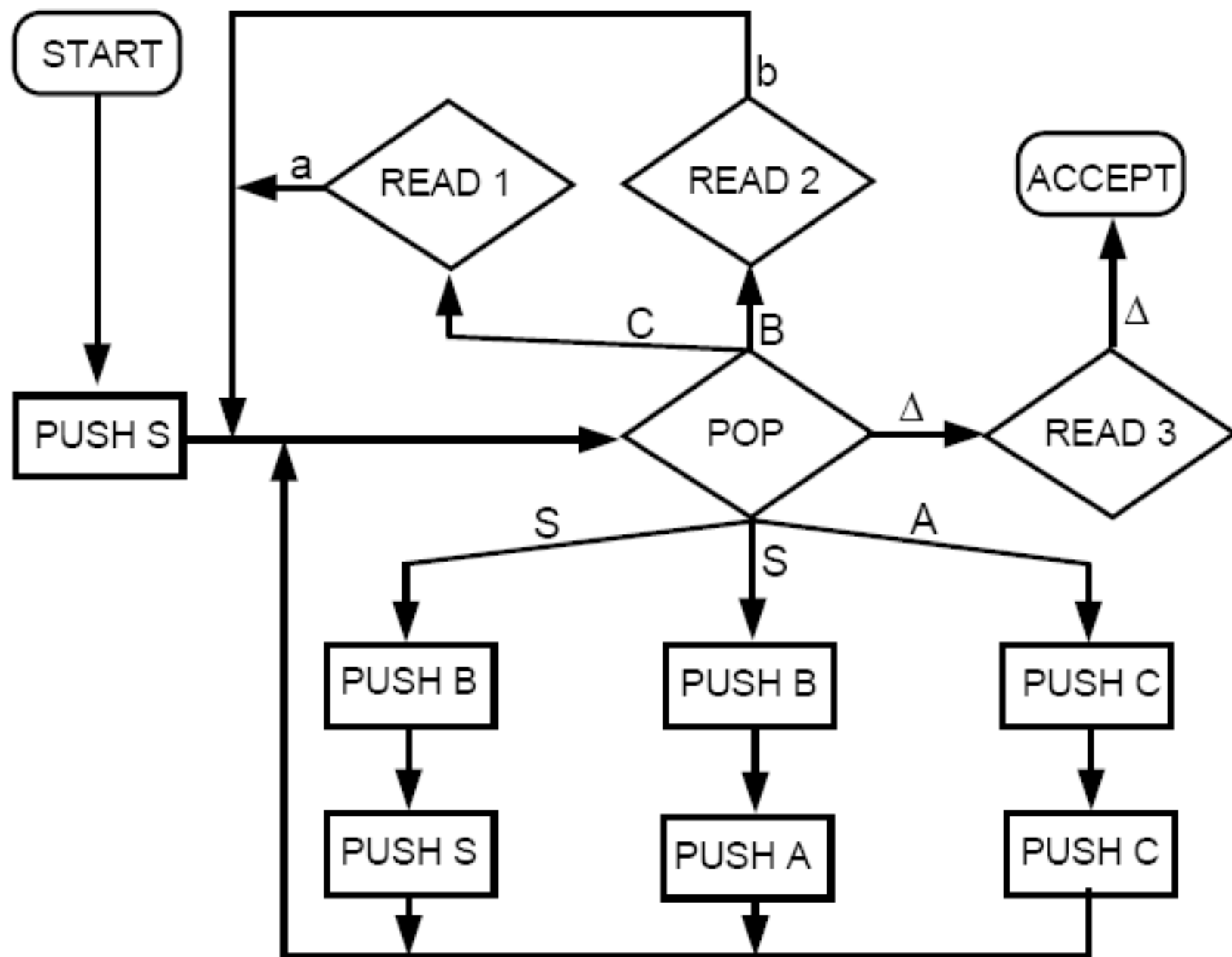
$$\Gamma = \{S, A, B, C\}$$

and the TAPE alphabet is only

$$\Sigma = \{a, b\}$$

Detail

- We begin by pushing S onto the top of the STACK.
- We then enter the central POP state. Two things are possible when we pop the top of the STACK:
 - We either replace the removed non-terminal with two other non-terminals, thereby simulating a production,
 - Or we go to a READ state, which insists that we must read a specific terminal from the TAPE, or else it crashes.
- To get to ACCEPT, we must have encountered the READ states that wanted to read exactly the letters on the INPUT TAPE.
- We now show that doing this is equivalent to simulating a leftmost derivation of the input string in the given CFG.



Construct aab string

- Let's consider a specific example. Let's generate the word ***aab*** using leftmost derivation in the given CFG:

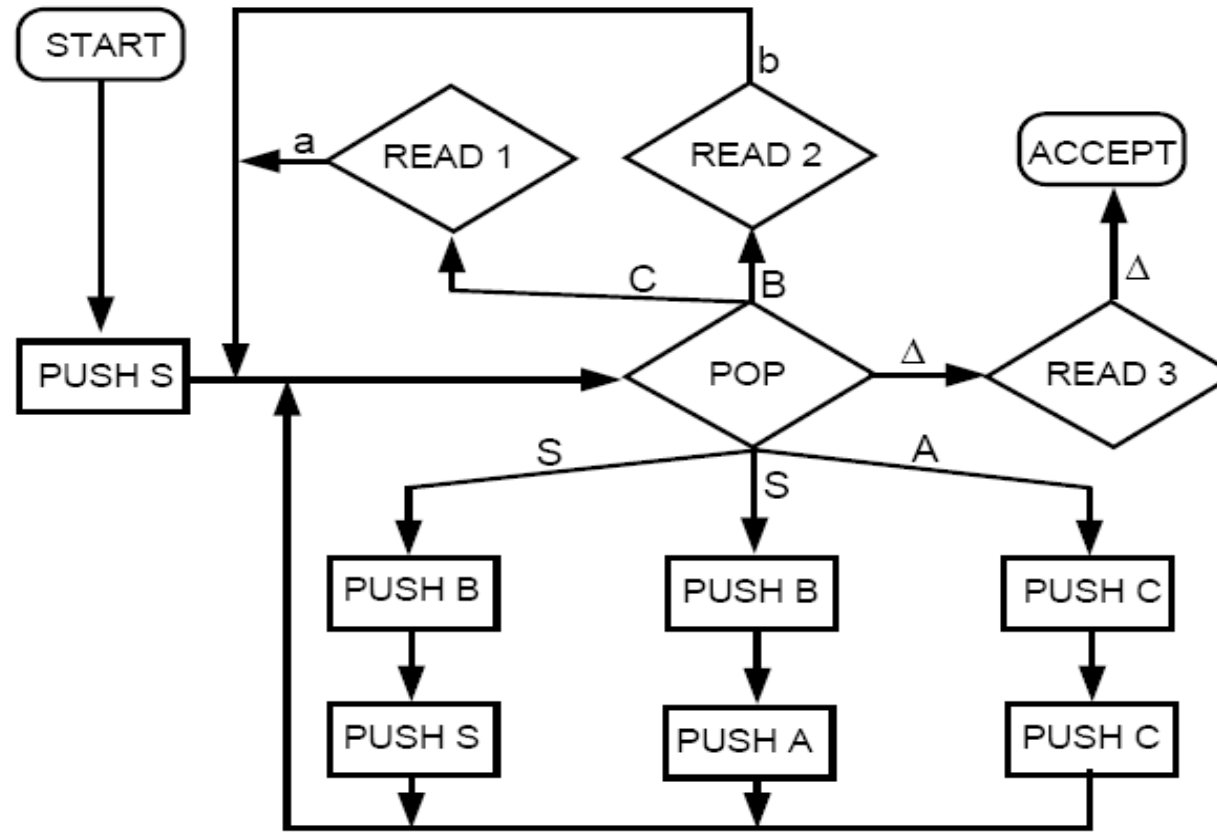
Working string

$S \rightarrow AB$
 $\rightarrow CCB$
 $\rightarrow aCB$
 $\rightarrow aaB$
 $\rightarrow aab$

Production used

$S \rightarrow AB$
 $A \rightarrow CC$
 $C \rightarrow a$
 $C \rightarrow a$
 $B \rightarrow b$

Let us run this word (***aab***) on the proposed PDA, following the same sequence of productions in the leftmost derivation above.



Example:

- We begin at START

STACK	TAPE
Δ	$aab\Delta$

- We push the symbol S on the STACK

STACK	TAPE
S	$aab\Delta$

- We then go to POP state. The first production we must simulate is $S \rightarrow AB$. So, we POP S and then PUSH B and PUSH A:

STACK	TAPE
AB	$aab\Delta$

Example:

- We go back to POP. We now simulate $A \rightarrow CC$ by popping A and do PUSH C and PUSH C :

STACK	TAPE
CCB	$aab\Delta$

- Again, we go back to POP. This time, we must simulate $C \rightarrow a$ by popping C and reading a from the TAPE:

STACK	TAPE
CB	$aab\Delta$

- We simulate another $C \rightarrow a$:

STACK	TAPE
B	$aaab\Delta$

Example:

- We now re-enter the POP state and simulate the last production, $B \rightarrow b$. We POP B and READ b from the TAPE

STACK	TAPE
Δ	<i>aab</i> Δ

- At this point the STACK is empty, and the blank Δ is the only thing we can read next from the TAPE.
- Hence, we follow the path
POP $\Delta \rightarrow$ READ3 $\Delta \rightarrow$ ACCEPT
- So, the word ***aab*** is accepted by the PDA.

Conclusion

- It should also be clear that if any input string reaches the ACCEPT state in the PDA, that string must have got there by having each of its letters read via simulating the Chomsky production of the form

Nonterminal $\rightarrow$ terminal

- This means that we have necessarily formed a complete leftmost derivation of this word through CFG productions with no nonterminals left over in the STACK. Therefore, every word accepted by this PDA is in the language generated by the CFG.

Example 2:

- The language PALINDROME (including λ) can be generated by the following CFG in CNF (plus one λ -production):

$$S \rightarrow AR_1 \mid BR_2 \mid AA \mid BB \mid a \mid b \mid \lambda$$

$$R_1 \rightarrow SA$$

$$R_2 \rightarrow SB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

- Using the algorithm above, we build the following PDA that accepts exactly the same language:

